

Enhanced Sampling for Coarse-Grained Machine Learning Potentials

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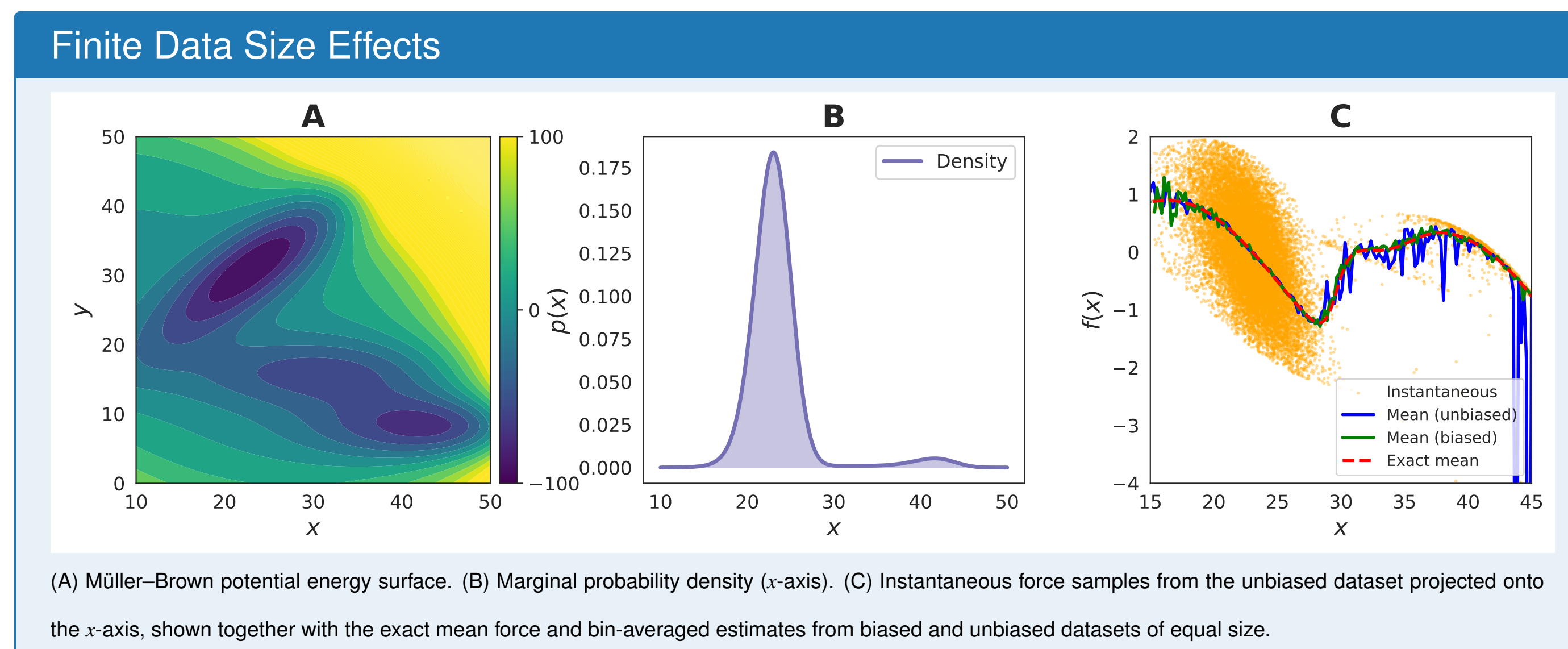
Motivation

Coarse-graining (CG) is essential for simulating large molecular systems over long timescales. Machine Learning Potentials (MLPs) are increasingly used to approximate the many-body Potential of Mean Force (PMF).

Problem: Standard "bottom-up" training (Force Matching) relies on unbiased MD simulations to generate training data (thermodynamic consistency).

- **Inefficiency** Requires prohibitively long trajectories to reach thermodynamic equilibrium.
- **Uneven Data Distribution** Samples cluster heavily in low-energy metastable basins and transition regions are rare.

Consequence: MLPs trained on unbiased MD data perform worse near transition regions, leading to incorrect thermodynamic statistics.



Unbiased Mean Forces from Biased Sampling

Key Theoretical Insight: When applying a bias potential $W(\mathbf{R})$ that depends only on the coarse-grained variables:

$$\mathbf{F}(\mathbf{R}) = \langle \xi(\mathbf{f}(\mathbf{r})) \rangle_{\mathbf{r}|\mathbf{R}} = \langle \xi(\mathbf{f}(\mathbf{r})) \rangle_{\mathbf{r}|\mathbf{W}|\mathbf{R}} \quad (1)$$

Proof: The bias changes how often a state is visited, but does not alter the relative probability of atomic microstates at that fixed coordinate. Since the bias $W(\mathbf{R})$ is constant at a fixed Coarse-Grained coordinate $\mathbf{R}$, it cancels out in the conditional probability:

$$p_W(\mathbf{r} | \mathbf{R}) = \frac{\delta(\xi(\mathbf{r}) - \mathbf{R}) e^{-\beta u(\mathbf{r})} \cdot e^{-\beta W(\mathbf{R})}}{\int \delta(\xi(\mathbf{r}) - \mathbf{R}) e^{-\beta u(\mathbf{r})} \cdot e^{-\beta W(\mathbf{R})} d\mathbf{r}} = p_{\text{unbiased}}(\mathbf{r} | \mathbf{R}) \quad (2)$$

Assumptions:

- **Bias Dependence:** W must depend exclusively on the CG variables $\xi(\mathbf{r})$.
- **Conditional Ergodicity:** Fast degrees of freedom (e.g., solvent) must equilibrate rapidly to the fixed CG.
- **Force Recovery** The bias force $-\nabla W$ must be subtracted from the sampled forces.

Proposed Workflow

1. Enhanced Data Generation

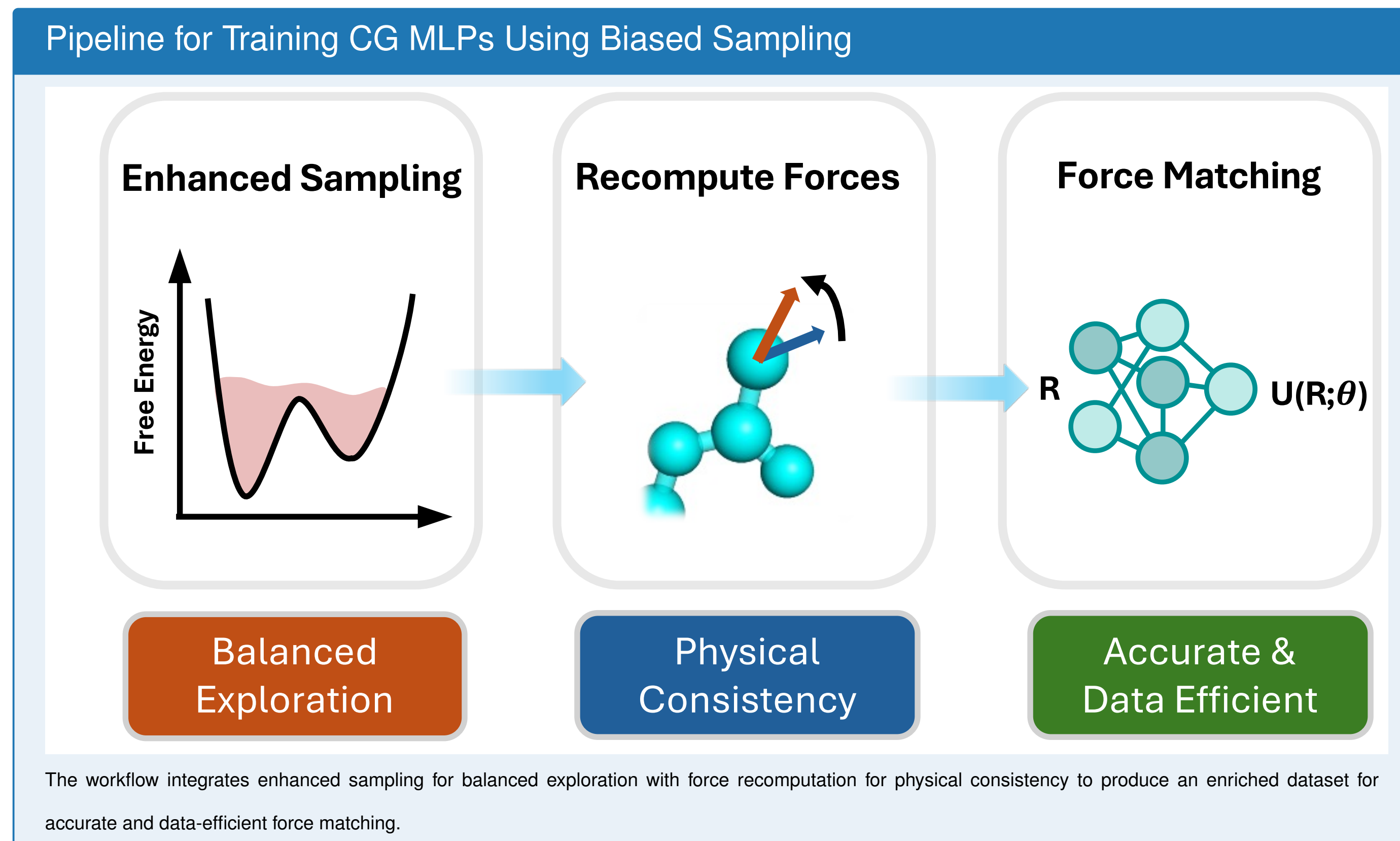
- Run Atomistic MD with Enhanced Sampling (e.g., Metadynamics, Umbrella Sampling) applied to the CG degrees of freedom.
- **Benefit:** Rapidly crosses energy barriers and explores the full configuration space.

2. Force Recomputation:

- For every sampled configuration, compute the forces using only the unbiased interaction potential (subtracting the bias force).

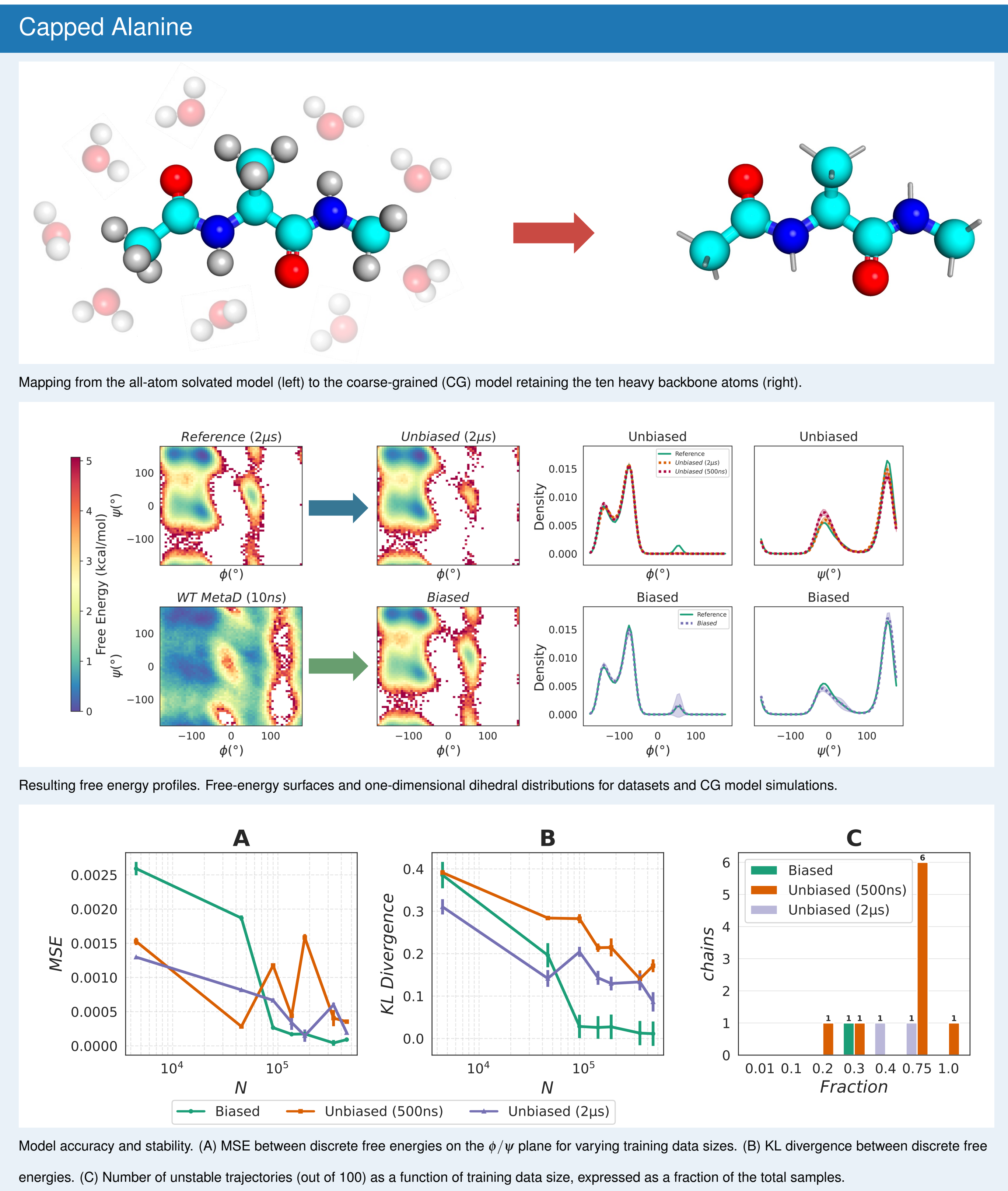
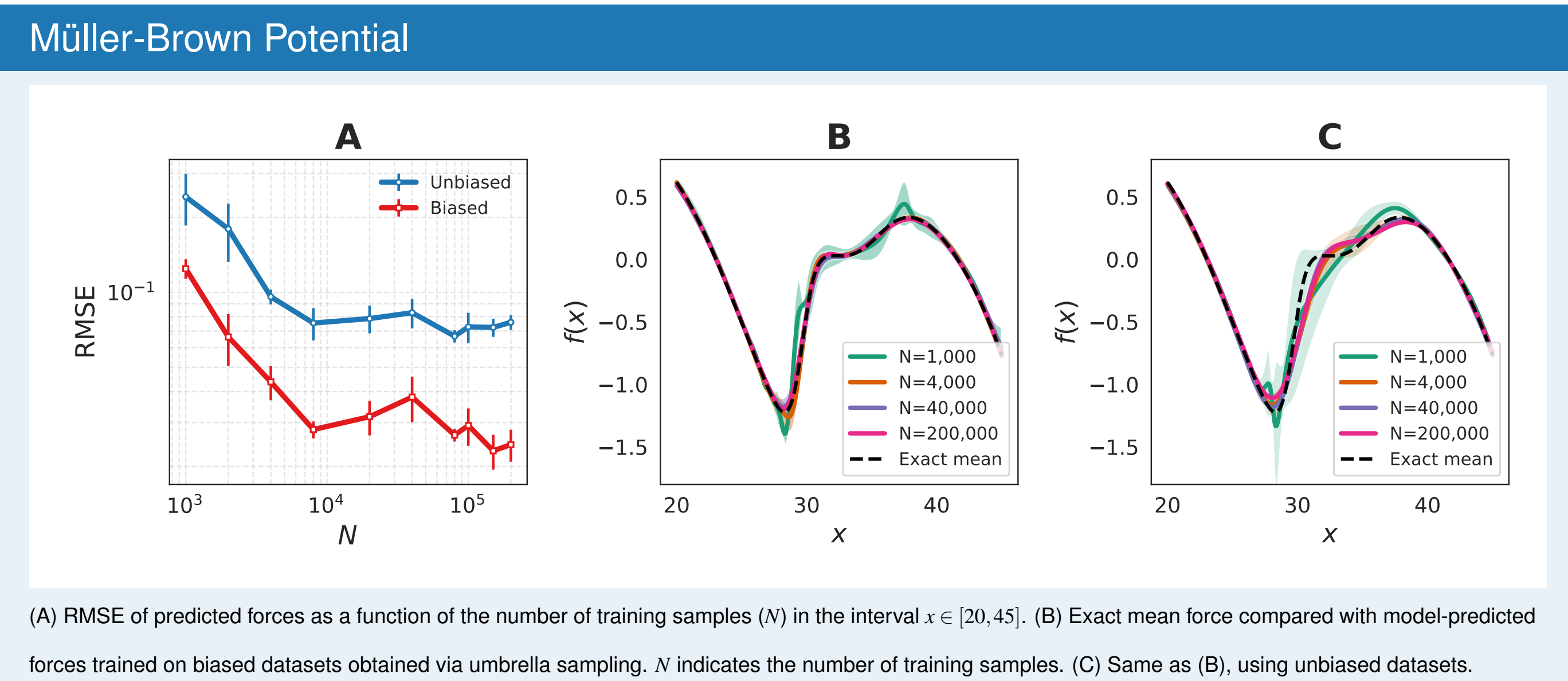
3. Training:

- Train the CG MLP using standard Variational Force Matching on this enriched dataset.



Experiments & Results

- **Müller-Brown Potential:** The 1D potential of mean force is approximated using a standard multi-layer perceptron with fully connected layers.
- **Capped Alanine:** The solvated peptide is modeled using the MACE architecture. Training is performed without empirical priors to evaluate the quality of the unbiased or biased data.



Advantages

- **Accelerated Convergence:** Enhanced sampling bypasses rare-event bottlenecks, drastically reducing the total simulation time required to generate dataset.
- **Higher Data Efficiency:** By driving the simulation to explore a broader and more diverse range of configurations, we achieve higher accuracy with fewer training points.
- **Superior Model Accuracy:** Our approach yields more accurate potentials, correctly resolving rare basins and high-energy regions where standard methods fail.

Outlook

- **Dependence on Collective Variables:** The current framework relies on pre-defined CVs for biasing. Future work will explore integrating automated CV discovery techniques to ensure efficient sampling of unknown landscapes.
- **Transferability:** Like most CG models, the learned potentials are specific to the training state. A key open challenge is extending this method to learn transferable potentials across different thermodynamic states and chemical spaces.
- **Towards Automated Pipelines:** Combine uncertainty quantification to create a closed-loop active learning pipeline. In this setup, the model's high-uncertainty regions would automatically drive the biasing, removing the need for manual setup.

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